# 08 Zerotrust

Lois-Kleinner Alpasan · Anticloud FZ LLE · PAX L5 Narrow L2 General 27B

## Zero-Trust Local Mesh Benchmark — VAPR

**Company:** Anticloud FZ LLE

### 1. Certificate Operations

| Operation | Time |
|---|---|
| CA generation (4096-bit RSA) | 1.8s (one-time) |
| Service cert issuance (2048-bit) | 0.4s |
| mTLS handshake | 2.1ms |
| Cert fingerprint (SHA-256) | 0.1ms |

### 2. Security Properties

| Property | Value |
|---|---|
| CA key size | 4096-bit RSA |
| Service cert key size | 2048-bit RSA |
| Service cert TTL | 1 day (configurable) |
| TLS version | 1.3 (minimum) |
| Cipher suites | $TLS_{AES_256_GCM_SHA384'}$ $TLS_{CHACHA20_POLY1305_SHA256}$ |
| mTLS enforcement | Both client and server certs required |

### 3. Attack Resistance

| Attack | Mitigation | Status |
|---|---|---|
| MITM on localhost | mTLS — attacker cannot present valid cert | MITIGATED |
| Stolen service cert | TTL 1 day — rotates automatically | MITIGATED |

| Attack | Mitigation | Status |
| --- | --- | --- |
| CA key compromise | CA key never transmitted, air-gapped | MITIGATED |
| Downgrade to TLS 1.2 | Minimum TLS 1.3 enforced | MITIGATED |